

Problem

Design an analog computer circuit to solve the differential equation:

$$\frac{d^2 v_o}{dt^2} + 3 \frac{dv_o}{dt} + 2v_o = 4 \cos 10t, \quad t > 0$$

subject to $v_o(0) = 2$, $v_o'(0) = 0$.

Step-by-step solution

Step 1 of 4

Write the differential equation.

$$\frac{d^2 v_o}{dt^2} + 3 \frac{dv_o}{dt} + 2v_o = 4 \cos 10t, \quad t > 0$$

Here, $v_o(0) = 2$, $v_o'(0) = 0$

Rearrange the expression for second derivative.

$$\frac{d^2 v_o}{dt^2} = 4 \cos 10t - 3 \frac{dv_o}{dt} - 2v_o$$

Integrate on both sides.

$$\int_0^t \frac{d^2 v_o}{dt^2} d\tau = \int_0^t \left[4 \cos 10\tau - 3 \frac{dv_o}{d\tau} - 2v_o \right] d\tau$$

$$\frac{dv_o}{dt} = - \int_0^t \left[-4 \cos(10\tau) + 3 \frac{dv_o(\tau)}{d\tau} + 2v_o(\tau) \right] d\tau + v_o'(0)$$

But here, the initial condition $\frac{dv_o(0)}{dt}$ is zero. So, the expression becomes,

$$\frac{dv_o}{dt} = - \int_0^t \left[-4 \cos(10\tau) + 3 \frac{dv_o(\tau)}{d\tau} + 2v_o(\tau) \right] d\tau$$

Step 2 of 4

Implementation of the circuit for $-\cos(10\tau)$ is shown in Figure 1.

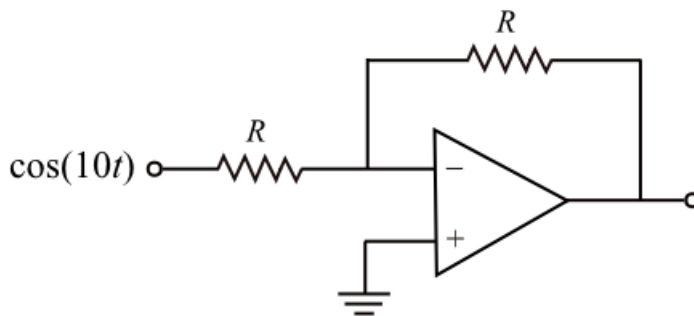


Figure 1

Step 3 of 4

The resistor and capacitor values have been chosen so that $RC = 1 \text{ s}$ for $-\frac{1}{RC} \int_0^t v_o(\tau) d\tau$.

Integrate $\frac{dv_o}{dt}$ to obtain the output voltage v_o .

$$v_o = -\int_0^t \left[-\frac{dv_o(\tau)}{dt} \right] d\tau + v(0)$$

Here, initial voltage is 2 V.

Implementation of the circuit for v_o is shown in Figure 2.

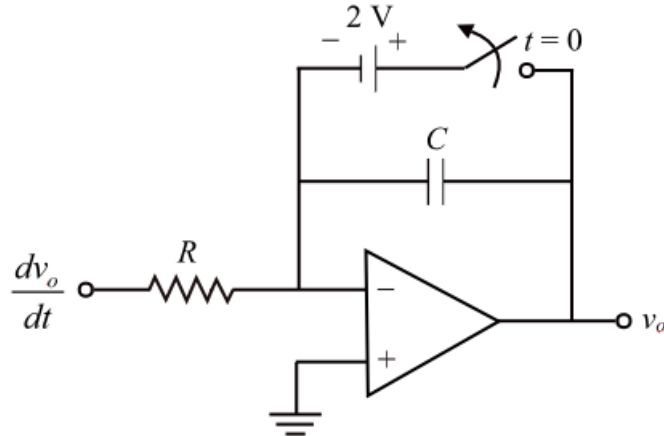


Figure 2

Step 4 of 4

Combine both the circuits of Figure 1 and Figure 2 to obtain the overall circuit diagram as shown in Figure 3.

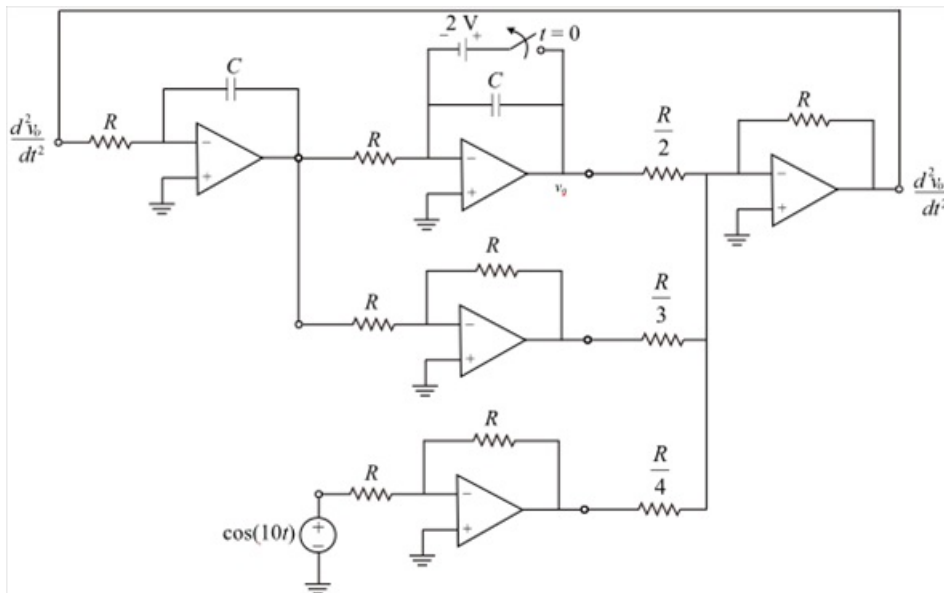


Figure 3

Thus, an analog computer circuit is designed to solve the differential equation.